



ALLIANCE
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Database Management Systems

AIRLINE RESERVATION SYSTEM

A Relational Database Design for Flight Booking & Management

***Bachelor of Computer Application
Semester – III***

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1. Introduction

An AirlineReservationSystem is a real-world database application used to manage flight reservations and related airline operations. The system allows passengers to search for flights, select a suitable flight, make a reservation, choose a seat, make payment, and receive a ticket.

A large amount of information is involved, including passenger details, flight schedules, airports, aircraft, seats, bookings, tickets, payments, and baggage. A relational database stores this information in separate but interconnected tables.

The database uses primary keys to uniquely identify records and foreign keys to establish relationships between related tables. This helps reduce duplication, maintain consistency, and efficiently retrieve information.

2. Name and Purpose of the Application

Application Name:AirlineReservationSystem

The purpose of an Airline Reservation System is to manage the process of booking and managing airline journeys through a database. It supports flight search, passenger information, seat selection, reservations, payments, ticket generation, baggage recording, and booking management.

3. Required Database Tables

No.	Table	Main Purpose
1	Passenger	Stores passenger information
2	Airport	Stores airport information
3	Aircraft	Stores aircraft details
4	Flight	Stores flight schedules and routes
5	Seat	Stores aircraft seat information
6	Booking	Stores flight reservations
7	Ticket	Stores issued ticket information
8	Payment	Stores payment details
9	Baggage	Stores baggage information

4. Database Tables and Attributes

4.1 Passenger Table

Attribute	Data Type	Key	Description
Passenger_ID	INT	PK	Unique passenger ID
First_Name	VARCHAR(50)	—	Passenger's first name
Last_Name	VARCHAR(50)	—	Passenger's last name
Email	VARCHAR(100)	—	Passenger's email
Phone	VARCHAR(15)	—	Passenger's phone number
Date_of_Birth	DATE	—	Passenger's date of birth
Gender	VARCHAR(10)	—	Passenger's gender

Primary Key: Passenger_ID

Purpose: Stores the personal information of passengers who book flights.

4.2 Airport Table

Attribute	Data Type	Key	Description
Airport_ID	INT	PK	Unique airport ID
Airport_Code	VARCHAR(10)	—	Airport code such as BLR
Airport_Name	VARCHAR(100)	—	Name of airport
City	VARCHAR(50)	—	City where airport is located
Country	VARCHAR(50)	—	Country
Terminal_Count	INT	—	Number of terminals

Primary Key: Airport_ID

Purpose: Stores information about airports used as flight departure and arrival locations.

4.3 Aircraft Table

Attribute	Data Type	Key	Description
Aircraft_ID	INT	PK	Unique aircraft ID
Aircraft_Model	VARCHAR(50)	—	Aircraft model
Registration_No	VARCHAR(30)	—	Aircraft registration number
Total_Seats	INT	—	Total seating capacity
Aircraft_Status	VARCHAR(20)	—	Active/Maintenance

Primary Key: Aircraft_ID

Purpose: Stores information about aircraft used to operate flights.

4.4 Flight Table

Attribute	Data Type	Key	Description
Flight_ID	INT	PK	Unique flight ID
Flight_Number	VARCHAR(20)	—	Airline flight number
Aircraft_ID	INT	FK	Aircraft assigned to flight
Departure_Airport_ID	INT	FK	Departure airport
Arrival_Airport_ID	INT	FK	Arrival airport
Departure_DateTime	DATETIME	—	Departure date and time
Arrival_DateTime	DATETIME	—	Arrival date and time
Flight_Status	VARCHAR(20)	—	Scheduled/Delayed/Cancelled

Primary Key: Flight_ID | **Foreign Keys:** Aircraft_ID, Departure_Airport_ID, Arrival_Airport_ID

Purpose: Stores flight number, aircraft, route, timings, and flight status.

4.5 Seat Table

Attribute	Data Type	Key	Description
Seat_ID	INT	PK	Unique seat ID
Aircraft_ID	INT	FK	Aircraft containing the seat
Seat_Number	VARCHAR(10)	—	Seat number such as 12A
Seat_Class	VARCHAR(20)	—	Economy/Business/First
Seat_Status	VARCHAR(20)	—	Available/Reserved

Primary Key: Seat_ID | **Foreign Key:** Aircraft_ID

Purpose: Stores seats available on each aircraft.

4.6 Booking Table

Attribute	Data Type	Key	Description
Booking_ID	INT	PK	Unique booking ID
Passenger_ID	INT	FK	Passenger making booking
Flight_ID	INT	FK	Booked flight
Seat_ID	INT	FK	Selected seat
Booking_Date	DATE VARCHAR(20)	—	Date booking was made
Booking_Status	DECIMAL(10,2)	—	Confirmed/Cancelled/Pending
Fare		—	Amount charged

Primary Key: Booking_ID | **Foreign Keys:** Passenger_ID, Flight_ID, Seat_ID

Purpose: Records a passenger's reservation for a flight and selected seat.

4.7 Ticket Table

Attribute	Data Type	Key	Description
Ticket_ID	INT	PK	Unique ticket ID
Booking_ID	INT	FK	Related booking
Ticket_Number	VARCHAR(30)	—	Ticket number
Issue_Date	DATE	—	Date ticket was issued
Ticket_Status	VARCHAR(20)	—	Valid/Cancelled

Primary Key: Ticket_ID | **Foreign Key:** Booking_ID

Purpose: Stores the ticket issued after a booking is confirmed.

4.8 Payment Table

Attribute	Data Type	Key	Description
Payment_ID	INT	PK	Unique payment ID
Booking_ID	INT	FK	Related booking
Payment_Date	DATE	—	Date payment was made

Attribute	Data Type	Key	Description
Payment_Method	VARCHAR(30)	—	Card/UPI/Net Banking
Amount	DECIMAL(10,2)	—	Amount paid
Payment_Status	VARCHAR(20)	—	Successful/Failed/Pending
Transaction_ID	VARCHAR(100)	—	Payment transaction reference

Primary Key: Payment_ID | **Foreign Key:** Booking_ID

Purpose: Stores payment information associated with a flight booking.

4.9 Baggage Table

Attribute	Data Type	Key	Description
Baggage_ID	INT	PK	Unique baggage ID
Booking_ID	INT	FK	Related booking
Baggage_Count	INT	—	Number of bags
Total_Weight	DECIMAL(5,2)	—	Total baggage weight
Excess_Baggage_Fee	DECIMAL(10,2)	—	Additional baggage charge

Primary Key: Baggage_ID | **Foreign Key:** Booking_ID

Purpose: Stores baggage-related information associated with a booking.

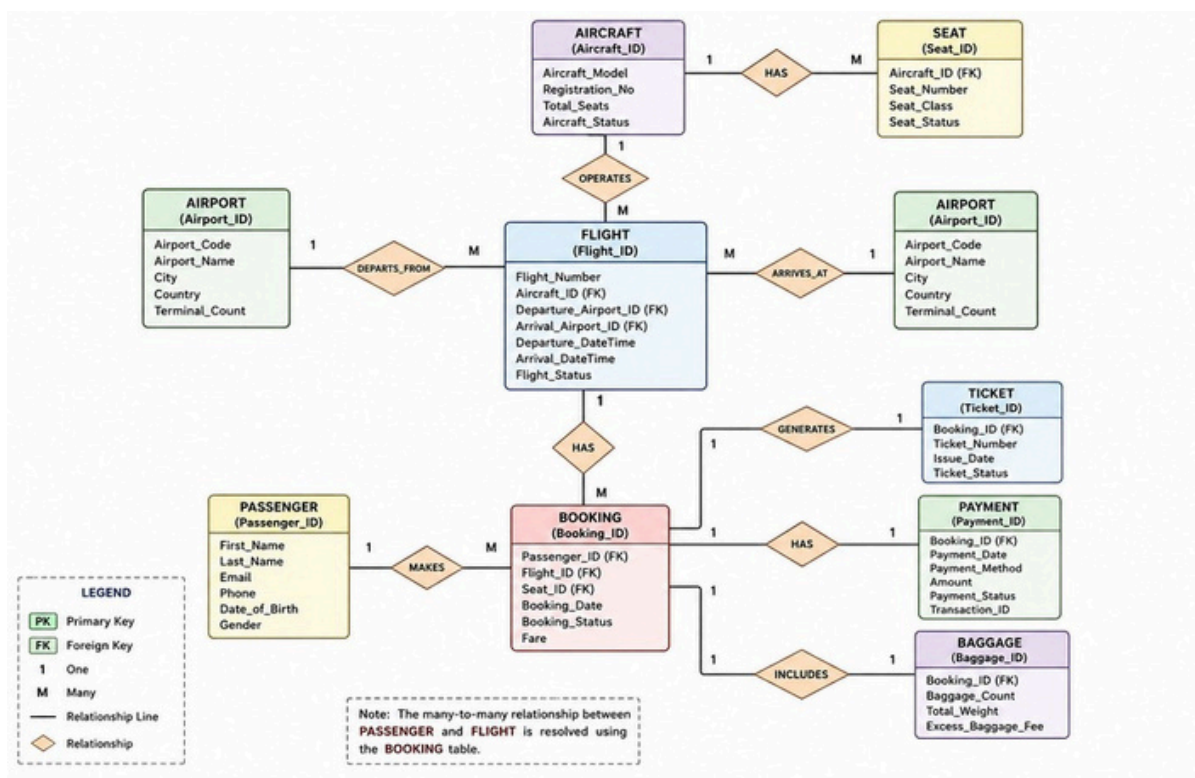
5. Relationships Between Tables

Relationship	Type	Explanation
Passenger → Booking	1 : M	One passenger can make multiple bookings
Airport → Flight	1 : M	One airport can be the departure point for many flights
Airport → Flight	1 : M	One airport can be the arrival point for many flights
Aircraft → Flight	1 : M	One aircraft can operate multiple flights over time
Aircraft → Seat	1 : M	One aircraft can operate multiple flights over time
Flight → Booking	1 : M	One aircraft contains many seats
Seat → Booking	1 : M	One flight can have many bookings
Booking → Ticket	1 : M	A seat can be booked across different flights
Booking → Payment	1 : 1	A confirmed booking can have one ticket
Booking → Baggage	1 : 1	A booking normally has one payment record
	1 : 1	A booking can have associated baggage information

Important DBMS concept: The relationship between Passenger and Flight is many-to-many because one passenger can book many flights and one flight can have many passengers. The Booking table resolves this relationship as an associative entity.

6. ER Diagram

Create the ER diagram using the tables and relationships defined above. The diagram should show the entities Passenger, Airport, Aircraft, Flight, Seat, Booking, Ticket, Payment, and Baggage, along with their primary keys, foreign keys, and cardinalities.



7. Working of the Airline Reservation System

1. Passenger Registration

The passenger provides personal information such as name, email, phone number, date of birth, and gender. This information is stored in the Passenger table.

2. Flight Search

The passenger searches for flights by selecting a departure location, destination, and travel date. The system retrieves suitable flights from the Flight table.

3. Flight Selection

The passenger selects a suitable flight. The Flight table provides the flight number, departure time, arrival time, aircraft, and flight status.

4. Seat Selection

The passenger selects an available seat. The Seat table stores the seat number, class, and availability.

5. Booking

Once the passenger confirms the flight and seat, a new record is created in the Booking table. The booking connects the passenger, flight, and selected seat.

6. Payment

The passenger makes payment using a supported payment method. The Payment table records the amount, method, transaction ID, and status.

7. Ticket Generation

After successful booking and payment, a ticket is generated. The Ticket table stores the ticket number and status.

8. Baggage Registration

If the passenger has baggage, the number of bags and total weight can be recorded in the Baggage table.

9. Booking Management

The passenger or airline can use the booking information to check booking status, ticket status, payment status, and other travel details.

8. Real-World Example

Consider a passenger booking a flight from Bengaluru to Delhi. The passenger searches for an available flight and selects a suitable flight. The Passenger table stores the passenger's personal details, while the Flight table stores the flight number, route, aircraft, and timings. When the passenger selects seat 12A, the Booking table connects the passenger with the selected flight and seat.

After the passenger makes a payment of ₹5,500 through UPI, the Payment table stores the payment information. Once the booking is confirmed, a ticket is generated and recorded in the Ticket table. If the passenger checks in baggage, the baggage count and weight are stored in the Baggage table.

9. Conclusion

The Airline Reservation System demonstrates how a real-world airline application can be represented using a relational database. The system divides information into meaningful tables such as Passenger, Airport, Aircraft, Flight, Seat, Booking, Ticket, Payment, and Baggage. Primary keys uniquely identify records, while foreign keys establish relationships between different tables. The Booking table plays an important role in connecting passengers with flights and resolving the many-to-many relationship between them. The database design helps reduce data duplication, maintain consistency, and efficiently manage flight reservations, passenger information, seat selection, payments, tickets, and baggage.